

ARUNIT BAIDYA

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EDUCATION

Northeastern University, Khoury College of Computer Sciences

Aug 2024 – April 2026

Master of Science, Computer Science

Boston, MA, US

Anna University, PSG College Of Technology

Aug 2018 – July 2022

Bachelor of Engineering, Computer Science and Engineering

Coimbatore, India

WORK EXPERIENCE

Amazon Web Services (Aurora) | *Databases, Distributed Systems, Backend, Cloud-Native*

Software Dev Engineer Intern

May 2025 - Aug 2025, Seattle

- **Migrated** single-instance concurrent R/W PostgreSQL **POC** changes in engine and filesystem to **newer releases**.
- Researched and implemented a **refactored pg_upgrade** major-upgrade sequence, **reducing downtime from several hours to < 1 minute**.
- Built **first of its kind test suite** to validate **major version upgrades** on local machines & **release pipelines**, catching dev-branch issues before promotion.

Arcesium | *Data, Backend, Cloud-Native*

Software Engineer

Jul 2022 - Jul 2024, Bangalore

- Profiled and optimized **Python/Java code** in distributed big data processing pipelines, **improving process times by 40%**.
- Built **JSON** config-driven data-filtering module for Spark ETL in **reducing compute usage by 30%** via partition pruning.
- Developed event-driven microservices with AWS SNS/SQS, **reducing latency by 25% through** batch size and timeout tuning.
- Optimized resource provisioning sizes (T-shirt model) in **ArgoWorkflow** and **Kubernetes**, **adopted in 10+ workflows**.
- Automated CI/CD pipelines with **Jenkins, Docker, & Kubernetes**, integrating JUnit for *testing ensuring 85% coverage*.
- Created LLD documentation with business logic translation, **reducing** cross-functional team **inquiries by 50%**.
- Actively participated in **Agile/SCRUM** teams to deliver **iterative enhancements** and optimize project workflows.

Software Engineering Intern

Feb 2022 - May 2022, Remote

- **Increased Spark ETL throughput by 70%** via preprocessing stage to split heterogeneous data for parallel processing.
- Enhanced system observability by integrating log publishing to **Elasticsearch** and creating **Datadog** monitors.

PROJECTS

Scalable Image Sharing Platform (GitHub) | *Golang, JavaScript, Kafka, Redis, Postgres, MinIO/S3*

- Architected a high-performance Golang backend utilizing **PostgreSQL** and **MinIO/S3** for scalable, reliable storage.
- Developed **RESTful APIs** for image uploads, likes, dislikes, views, and comments with **batched database updates**.
- Leveraged **Kafka** for event-driven processing and **Redis** caching (TTL / LRU eviction) improving request response times by **40%**.
- Deployed using **Docker**, with **JWT authentication, rate limiting, & Prometheus** monitoring for enhanced observability.

Image Processing Application (GitHub) | *Java, SOLID, Object-oriented, MVC*

- Developed an image processing application in **Java** supporting 10+ operations, including blur, compress, and downscale, with execution options via terminal, script, or GUI inputs.
- Used SOLID principles, MVC architecture, and **design patterns** (Command, Factory etc.), **reducing code complexity by 40%**.

egOS (GitHub) | *C, Operating Systems, RISC V, Multithreaded Programming*

- Enhanced egOS by implementing **multilevel feedback queue** (MLFQ) kernel scheduling, **user-level threading**, **virtual memory management** via paging, and **physical memory protection** on RISC-V architecture in **C**.

SKILLS

Programming Languages:	Python, Java, Go, C, JavaScript, HTML, CSS
Frameworks and Libraries:	REST, Redis, Swing, Pytest, JUnit, Argo Workflow, Ansible, Prometheus
Data Engineering:	Spark, Kafka, ETL, DeltaLake, Snowflake, PostgresDB, MySQL, NoSQL, RDBMS
Cloud and DevOps:	REST API development, AWS, Elasticsearch, SNS, SQS, Terraform, Docker, Kubernetes
Software Development:	Data Structures & Algorithms, Agile Development, Test-Driven Development, Code Reviews, Git